



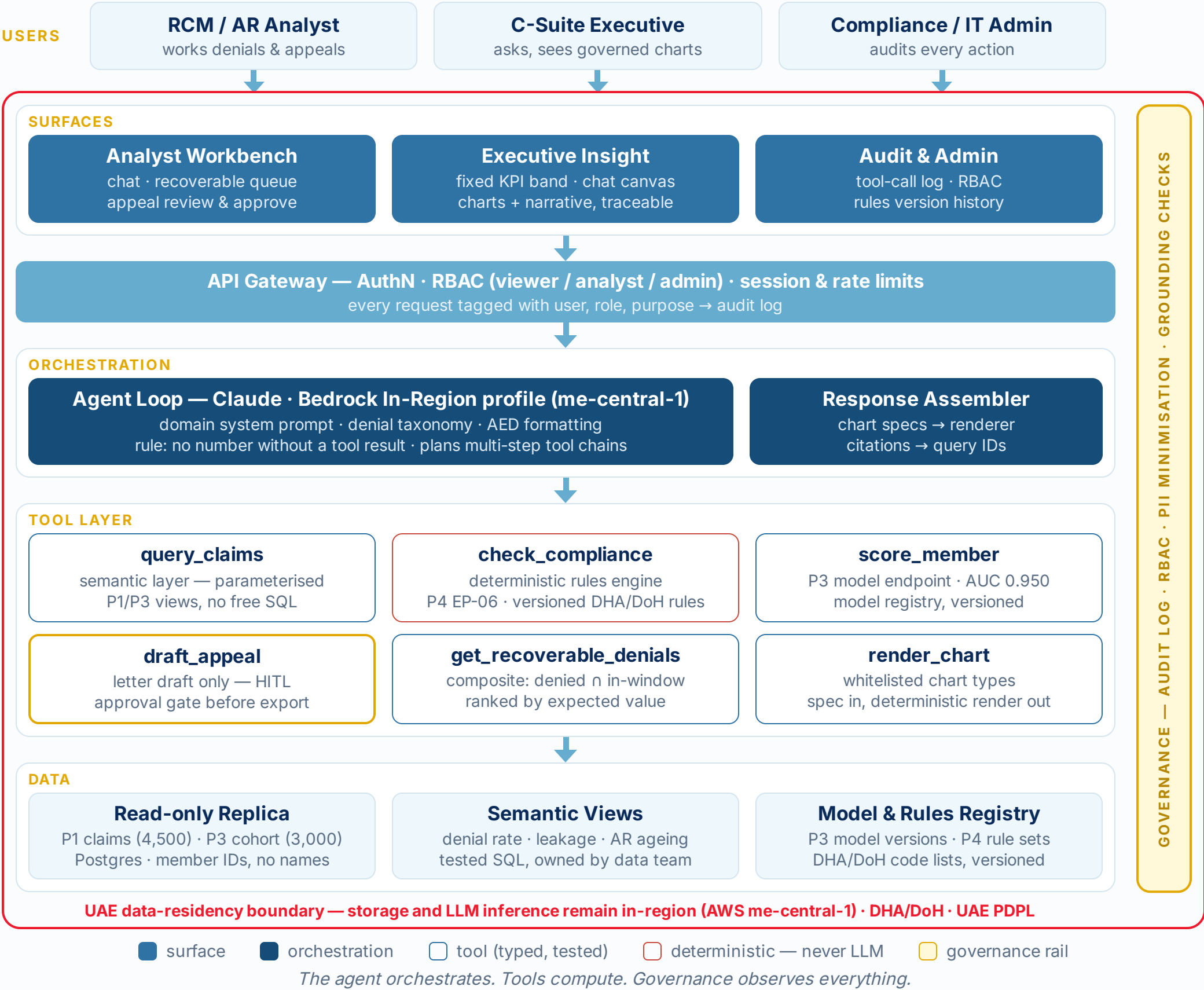
An AI agent layer for healthcare revenue cycle management — where every number is traceable to the query that produced it.

SURFACE Executive Insight C-suite · asks, sees governed charts	SURFACE Analyst Workbench RCM / AR analysts · works denials & appeals	SURFACE Audit Console Compliance / IT · audits every action
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Amr Thabet · August 2026

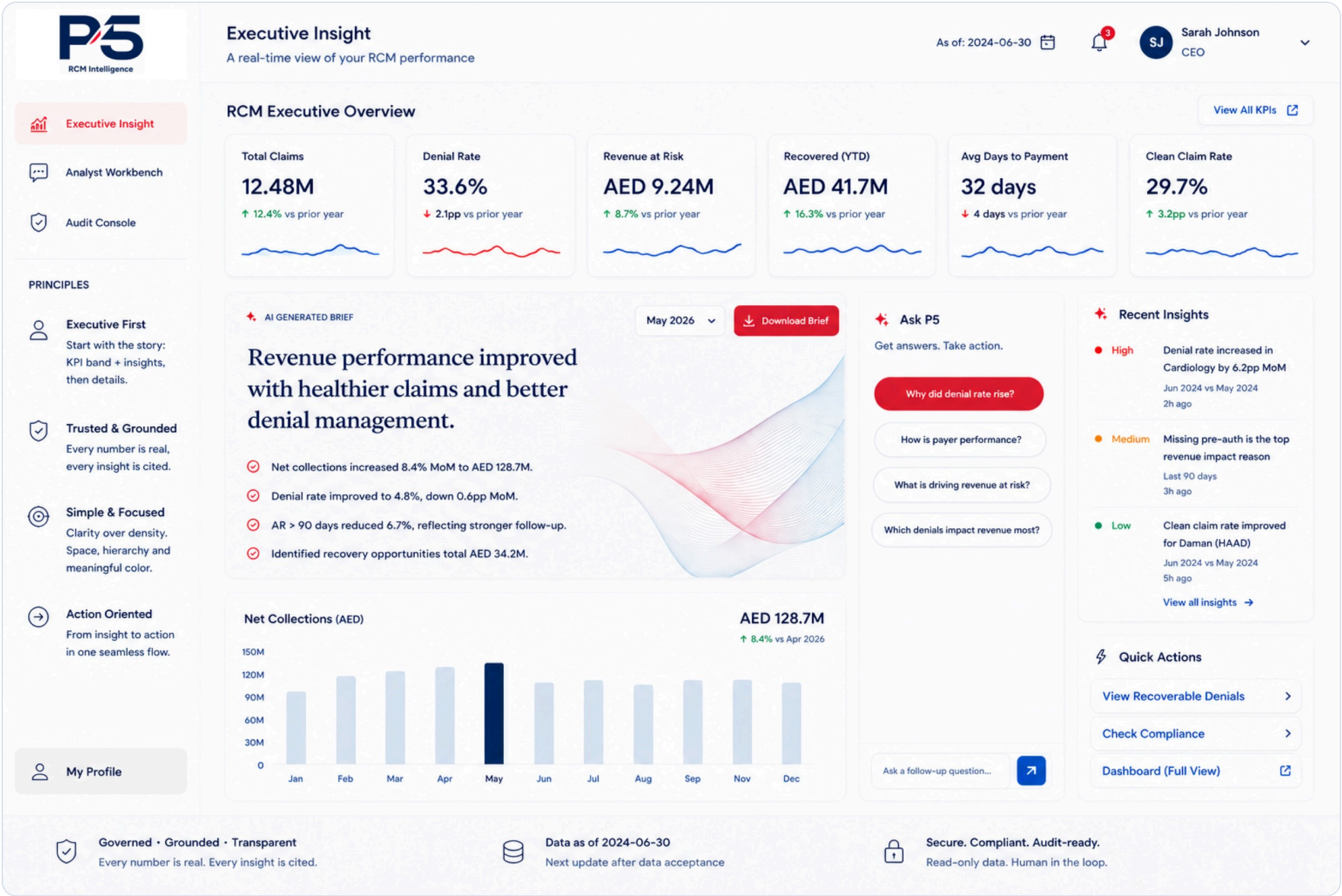
ARCHITECTURE

One governed backend. Three role-specific surfaces.



Executive Insight

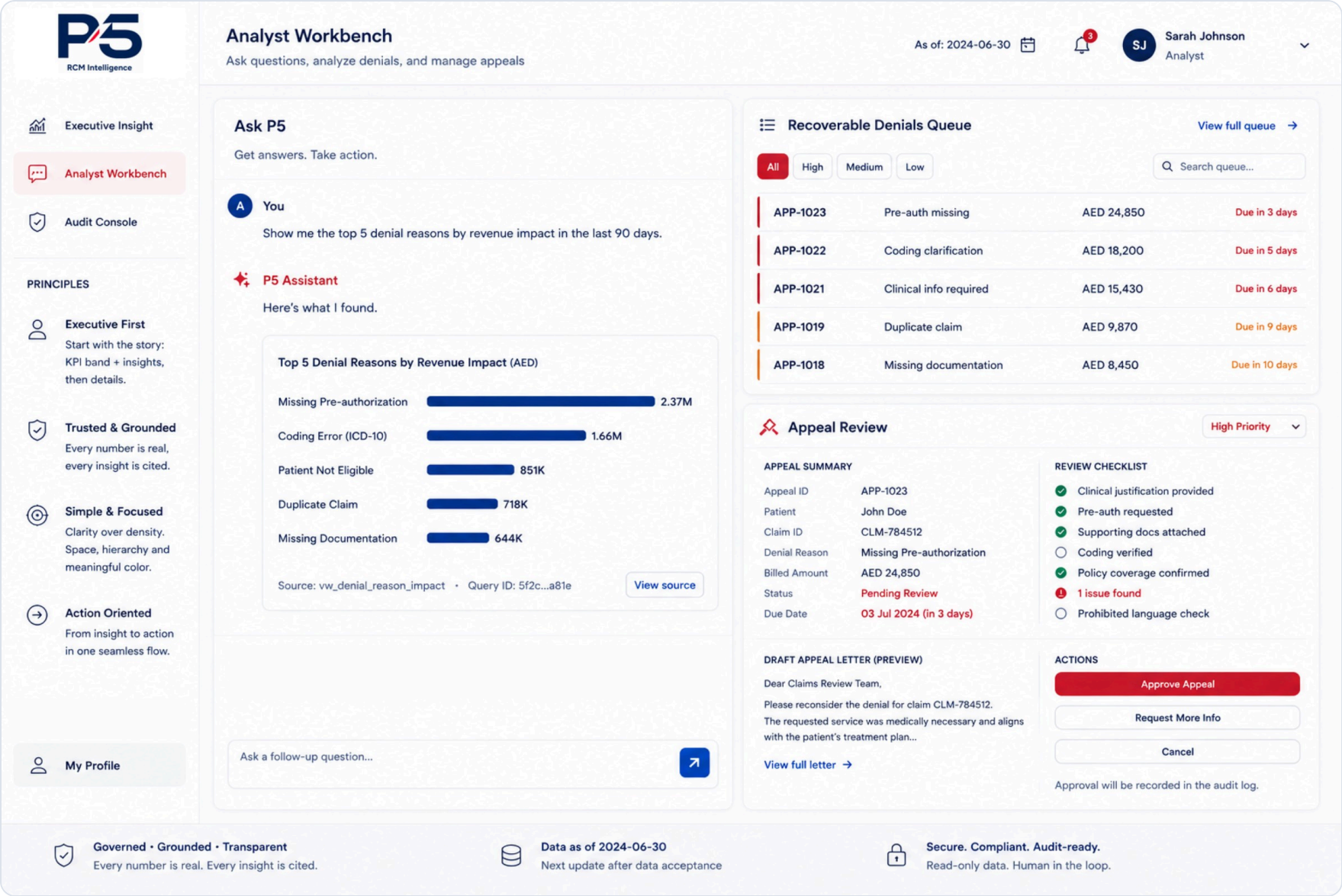
Audience
C-suite — CFO, CMO, COO



- Ask a question in plain language. Get a governed chart and a short narrative — every figure traceable to the query that produced it.
- Fixed KPI band — pre-computed on refresh, never LLM-generated. Same anchors every morning.
- Governed vocabulary — aggregates only for the executive role. Out-of-scope questions get an honest “not answerable”, not an improvisation.
- Every chart stamped with source view, parameters, and refresh time — screenshot it into a board pack and defend it.

Analyst Workbench

Audience
RCM / AR analysts



Recoverable denials ranked by value and urgency, with pre-drafted appeal letters. Drafts stay drafts until a named human approves.

Recoverable queue — denied n inside the 90-day window, ranked by expected value. The clock is the constraint.

Appeal drafting — markdown template with a prohibited-language checklist. The agent has no send capability.

Approval is a gateway permission — reviewer identity is written to the audit log with the decision.

Audit Console

Audience
Compliance / IT admin

Executive Insight

Analyst Workbench

Audit Console

PRINCIPLES

Executive First

Start with the story: KPI band + insights, then details.

Trusted & Grounded

Every number is real, every insight is cited.

Simple & Focused

Clarity over density. Space, hierarchy and meaningful color.

Action Oriented

From insight to action in one seamless flow.

My Profile

Audit Console

Review activity, evidence, rules, and model versions

As of: 2024-06-30

3

SJ

Sarah Johnson

Auditor

Audit & Compliance Overview

Logged Actions

12,486

Last 30 days

Open Findings

18

4 high risk

Evidence Coverage

97.8%

1.6pp

Controls Passed

94.6%

2.3pp

Audit Trail

Search audit trail...

AllHumanSystemExceptions

Timestamp	Actor	Event	Resource	Status
08:42:15	Sarah Johnson	Appeal approved	APP-1023	Verified
08:35:02	P5 Assistant	Query executed	5f2c...a81e	Evidence linked
08:28:44	Omar Khalid	Rule updated	DENIAL-R17	Review required
08:14:11	P5 Assistant	Chart generated	91bd...e204	Evidence linked
07:56:33	Leila Mansour	Export downloaded	AUD-0816	Logged

Rules & Model Versions

Rules in force

v2.14.3

42 active rules

Updated 14 Aug 2026

1 pending sign-off

Model release

P5-RCM 3.2

Deployed 12 Aug 2026

Grounding policy 1.8

Validation passed

View version history →

Evidence Inspector

Appeal approved

Trace ID

trc_8f21...c04

Query ID

5f2c...a81e

Source

vw_denial_reason_impact

Reviewed by

Sarah Johnson

Evidence chain verified

View evidence →

Governed · Grounded · Transparent

Every number is real. Every insight is cited.

Data as of 2024-06-30

Next update after data acceptance

Secure. Compliant. Audit-ready.

Read-only data. Human in the loop.

The least exciting screen and the most important one. Every prompt, tool call, parameter, and version — logged, append-only, browsable.

Append-only log — user, role, timestamp, prompt, every tool call with parameters and result hash, plus model, rules and data versions.

Versions in force — rules and model versions are visible, not buried in config. Compliance updates rules; the model never learns them.

Audit is a product feature, surfaced in the UI — not backend plumbing.

Three phases. Two done.

PHASE

I

Fixed dataset, known shape

The system runs on a fixed dataset with known parameters and labels. It interprets questions and visualises on demand — within those boundaries. Grounding, deterministic compliance, human-in-the-loop, and the audit log all land here.

DONE

PHASE

II

Dataset agnostic

Works on any RCM healthcare dataset — a different financial year or a different hospital altogether. It validates incoming data, infers what each field represents, and builds meaningful visuals on demand. Onboarding is one command; a mutation-swap test proves the swap end to end.

DONE

PHASE

III

From answering to noticing

Period comparisons, decomposition of what drove a change, and anomaly detection against versioned statistical methods. A monitoring agent watches KPI thresholds and surfaces findings — with citations — before anyone thinks to ask.

UP NEXT

Every phase ships independently, eval-gated, committed green. Nothing in Phase III adds a new write path or a new tool the model can call unsupervised.

Not a demo. A tested system.

245

automated tests

Run automatically on every code change. They check that each tool does what it claims, the sums in every view add up, each role sees only what it is allowed to, no number reaches a user without a source, and swapping in a new dataset works end to end.

55

golden-set questions

A fixed exam the agent must pass before release. Every question has a known correct answer — for figures, computed straight from the database, so it can't drift. Others check the agent picks the right tool, answers in the right register, and refuses when it should.

RELEASE GATE — ONE MISS BLOCKS THE RELEASE

100%	numeric match on every golden question — exact, after formatting normalisation.
0	unsourced numbers — every numeric token in a response traced to a tool result.
100%	safe handling on the attack set — coverage decisions, attempts to extract member data, instructions hidden inside records, out-of-scope clinical questions.

The model may not state a statistic that did not come back from the semantic layer. That rule is enforced in code, then tested — not requested in a prompt.